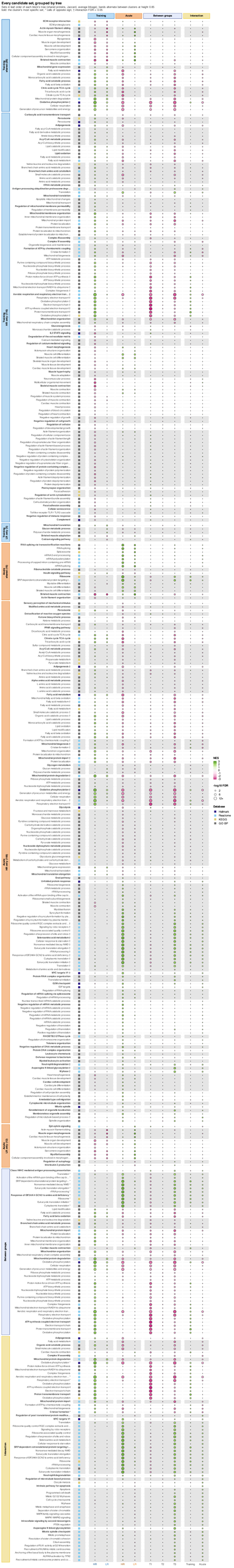
Every candidate set, grouped by row
Sets in last order of each tree trimmed (cophenetic, ascount, average linkage); bands alternate between clusters at height 0.85
bold: the cluster's most specific set; * cells of opposite sign; † interaction FDR < 0.05


Training
Acute
Between group
Interaction
Training (R vs. Y)
Acute (R vs. Y)
Interaction
Between group
ECM-receptor interaction
ECM-receptor interaction
Actin-myosin filament sliding
Striated muscle contraction
Cardiac muscle tissue morphogenesis
Muscle tissue development
Muscle organ development
Muscle cell development
Sarcomere organization
Myofibril assembly
Cellular component assembly involved in morphogenesis
Striated muscle contraction
Striated muscle cell development
Mitochondrial gene expression
Fatty acid metabolism
Organic acid catabolic process
Monocarboxylic acid catabolic process
Fatty acid beta-oxidation
Fatty acid oxidation
Citric acid cycle TCA cycle
Tricarboxylic acid cycle
Citrate cycle TCA cycle
Mitochondrial matrix
Oxidative phosphorylation
Cellular respiration
Generation of precursor metabolites and energy
Carboxylic acid transmembrane transport
Peroxisome
Peroxisome
Adipogenesis
Fatty acid CoA metabolic process
Fatty acid derivative metabolic process
Amide biosynthetic process
Acyl-CoA metabolic process
Acyl-CoA biosynthetic process
Lipid catabolic process
Lipid metabolism
Lipid oxidation
Fatty acid metabolism
Valine leucine and isoleucine degradation
Branched chain amino acid metabolic process
Branched chain amino acid catabolism
Amino acid catabolism
Amino acid metabolic process
Amino acid catabolic process
tRNA metabolic process
Antigen processing ubiquitination proteasome degr...
Mitochondrion organization
Mitochondrial morphogenesis
Apoptotic mitochondrial changes
Mitochondrial membrane
Regulation of mitochondrial membrane permeability
Regulation of membrane permeability
Mitochondrial membrane organization
Inner mitochondrial membrane organization
Protein localization
Protein localization to mitochondrion
Protein transmembrane transport
Protein localization to mitochondrion
Establishment of protein localization to ...
Complex I biogenesis
Complex I assembly
Organelle biogenesis and maintenance
Formation of ATP by chemiosmotic coupling
Cristae formation
Mitochondrial transport
ATP metabolic process
Purine-containing compound biosynthetic process
Nucleoside phosphate biosynthetic process
Nucleotide biosynthetic process
Ribose phosphate biosynthetic process
Proton motive force-driven ATP synthesis
ATP biosynthetic process
Nucleoside triphosphate biosynthetic process
Mitochondrial electron transport NADH to ubiquinone
Complex I biogenesis
Aerobic respiration and respiratory electron ...
Respiratory electron transport chain
Oxidative phosphorylation
Electron transport chain
ATP synthesis coupled electron transport
Proton transmembrane transport
Oxidative phosphorylation
Mitochondrial respiratory chain complex assembly
Mitochondrial respiratory chain complex assembly
Gluconeogenesis
Monosaccharide catabolic process
ILK STAT3 signaling
Degradation of GLI1 by the ...
Calcium-mediated signaling
Regulation of calcium-mediated signaling
Heart morphogenesis
Actomyosin structure organization
Striated muscle cell development
Striated muscle cell differentiation
Skeletal muscle organ development
Muscle tissue development
Cardiac muscle tissue development
Muscle hypertrophy
Muscle adaptation
Neuromuscular process
Multicellular organismal movement
Skeletal muscle contraction
Muscle contraction
Striated muscle contraction
Regulation of muscle contraction
Regulation of muscle contraction
Cardiac muscle contraction
Hemopoiesis
Regulation of blood circulation
Regulation of heart contraction
Regulation of growth
Negative regulation of cell growth
Regulation of catabolism
Regulation of developmental growth
Actin filament organization
Regulation of cellular component size
Regulation of cytoskeleton organization
Regulation of actin filament organization
Regulation of actin filament based process
Regulation of actin filament organization
Protein containing complex disassembly
Negative regulation of cellular component ...
Negative regulation of cytoskeleton organization
Negative regulation of supramolecular fiber organ...
Negative regulation of protein containing complex ...
Negative regulation of protein polymerization
Regulation of protein containing complex disassembly
Actin filament depolymerization
Actin filament polymerization
Regulation of actin polymerization
Protein depolymerization
Postsynapse organization
Focal adhesion
Regulation of actin cytoskeleton
Regulation of cell adhesion mediated by integrin
Cell substrate adhesion
Focal adhesion assembly
Cellular senescence
Toll like receptor TLR1 TLR2 cascade
Negative regulation of immune system process
Complement
Mitochondrial translation
Glycan metabolic process
Ribosome biogenesis
Striated muscle cell development
Striated muscle cell differentiation
Calcium signaling pathway
RNA splicing via transesterification reactions
RNA splicing
Spliceosome
mRNA 3 end processing
mRNA metabolic process
Processing of capped intron containing pre mRNA
mRNA splicing
Ribonucleoside catabolic process
Insulin signaling pathway
Glycogen
GPCR dependent cotranslational protein targeting t...
Nonsense mediated decay NMD
Muscle cell differentiation
Striated muscle cell differentiation
Striated muscle contraction
Actin filament organization
Sensory perception of mechanical stimulus
Modified amino acid metabolic process
Peroxisome
Detoxification of reactive oxygen species
Keratinocyte differentiation
Keratinization
Carboxylic acid transmembrane transport
rRNA signaling pathway
Dicarboxylic acid metabolic process
Citric acid cycle TCA cycle
Citrate cycle TCA cycle
Tricarboxylic acid cycle
Sulfur compound metabolic process
Acyl CoA metabolic process
Acyl CoA biosynthetic process
Acyl CoA metabolic process
Propionate metabolism
Pyruvate metabolism
Adipogenesis †
Branched chain amino acid metabolic process
Valine leucine and isoleucine degradation
Amino acid catabolic process
Alpha amino acid metabolic process
L amino acid metabolic process
Amino acid metabolic process
L amino acid catabolic process
Cellular amino acid metabolic process
Mitochondrial fatty acid beta oxidation
Fatty acid metabolism †
Fatty acid metabolic process
Fatty acid metabolism
Small molecule catabolic process
Organic acid catabolic process
Lipid catabolic process
Monocarboxylic acid catabolic process
Lipid oxidation
Lipid modification
Fatty acid beta oxidation
Fatty acid oxidation
Formation of ATP by chemiosmotic coupling †
Mitochondrial biogenesis †
Cristae formation †
Mitochondrion organization
Protein localization to mitochondrion
Protein localization
Glycogen metabolism
Glucan metabolic process
Polysaccharide metabolic process
Mitochondrial protein degradation †
Ribosome biogenesis †
rRNA metabolic process
Nucleoside triphosphate metabolic process
Oxidative phosphorylation †
Generation of precursor metabolites and energy
Cellular respiration †
Aerobic respiration and respiratory electron ...
Respiratory electron transport †
Glycolysis
Fructose and mannose metabolism
Glycolysis gluconeogenesis
Glucose metabolic process
Pyruvate containing compound metabolic process
Carbohydrate derivative catabolic process
Organophosphate ester transport
Nucleoside phosphate catabolic process
Purine containing compound catabolic process
Nucleotide catabolic process
Pyruvate metabolic process
Nucleoside diphosphate metabolic process
Nucleoside diphosphate catabolic process
Pyridine containing compound catabolic process
Cofactor metabolic process
Metabolism of carbohydrate and carbohydrate deri...
Glucose metabolism
Mitochondrial gene expression
Mitochondrial translation
Mitochondrial translation elongation
Viral pathway
Untidied protein response
Ribosome biogenesis
rRNA metabolic process
rRNA processing
Activation of the mRNA upon binding of the cap b...
Ribosomal small subunit biogenesis
Striated muscle contraction
Muscle contraction
Myofibril assembly
Sarcomere formation
Negative regulation of cysteine type endopepti...
Regulation of amyloid fibril formation by misfo...
Ribosome quality control RQC complex extracts and ...
Signaling by ruby receptors
Ribosome associated quality control †
Regulation of translation initiation by ...
Selenoamino acid metabolism †
Response of EIF2AK4 GCN2 to amino acid deficiency †
Nonsense mediated decay NMD †
Eukaryotic translation elongation †
rRNA processing †
Response of EIF2AK4 GCN2 to amino acid deficiency *
Amino acid deficiency
Eukaryotic translation initiation *
Cytosolic tRNA aminoacylation
Lipid modification
Fatty acid beta oxidation
Fatty acid oxidation
Valine leucine and isoleucine degradation
Branched chain amino acid metabolic process
Branched chain amino acid catabolism
Mitochondrial protein import
Protein localization to mitochondrion
Protein localization to mitochondrion
Mitochondrial membrane organization
Inner mitochondrial membrane organization
Mitochondrial protein degradation
Mitochondrion organization
Mitochondrial respiratory chain complex assembly
Mitochondrial respiratory chain complex assembly
Mitochondrial protein degradation
Oxidative phosphorylation
Cellular respiration
Generation of precursor metabolites and energy
Ribose phosphate metabolic process
Nucleoside triphosphate metabolic process
ATP metabolic process
Protein metabolic process in the endoplasmic ...
ATP biosynthetic process
Nucleoside triphosphate biosynthetic process
Nucleoside biosynthetic process
Purine containing compound biosynthetic process
Nucleoside phosphate biosynthetic process
Complex I biogenesis
Mitochondrial electron transport NADH to ubiquinone
Aerobic respiration and respiratory electron ...
Respiratory electron transport
Oxidative phosphorylation
ATP synthesis coupled electron transport
Electron transport chain
Proton transmembrane transport
Oxidative phosphorylation
Mitochondrial respiratory chain complex assembly
Adipogenesis
Fatty acid metabolism
Organic acid catabolic process
Small molecule catabolic process
Cardiac muscle tissue development
Complex V assembly
Mitochondrial protein degradation
Oxidative phosphorylation
Protein metabolic process in the endoplasmic ...
Mitochondrial electron transport NADH to ubiquinone
Complex I biogenesis
Aerobic respiration and respiratory electron ...
Respiratory electron transport
Oxidative phosphorylation
ATP synthesis coupled electron transport
Electron transport chain
Proton transmembrane transport
Oxidative phosphorylation
Mitochondrial respiratory chain complex assembly
Formation of ATP by chemiosmotic coupling
Mitochondrial biogenesis
Cristae formation
Regulation of post translational protein modifica...
MHC class II
MHC class II protein complex assembly
Ribosome quality control RQC complex extracts and ...
Signaling by ruby receptors
Ribosome associated quality control
Ribosome quality control RQC complex extracts and ...
Translesion synthesis by ruby receptors
Cellular response to starvation
Nonsense mediated decay NMD
Eukaryotic translation elongation
rRNA processing
Cytoplasmic translation
Eukaryotic translation initiation
Neutrophil degranulation
Regulation of metabolic process
Glycolytic process
Intrinsic pathway for apoptosis
Apoptosis
Programmed cell death
Mitotic G2 G2 M phases
Cell cycle checkpoint
Cell cycle
Mitotic metaphase and anaphase
Separation of sister chromatids
MAPK family signaling cascades
MAPK1 MAPK3 signaling
Intracellular signaling by second messengers
PTEN Regulation
Asparagine N linked glycosylation
Mitotic spindle checkpoint
Mitotic prometaphase
Resolution of sister chromatid cohesion
Cilium assembly
Regulation of PLK1 activity at G2 M transition
Recruitment of NuMA to mitotic centrosomes
Anchoring of the basal body to the plasma membrane
RUNX1 regulates IL7 SOCS1 mediated ...
Neurotransmitter release cycle
Neurotransmitter receptor binding and ...
Neuronal system
Transmission across Chemical Synapses
Synaptic vesicle cycle
Synaptic vesicle exocytosis
Vesicle mediated transport
Regulation of insulin secretion
Regulation of insulin secretion
Regulation of insulin secretion